Basics of automation technology

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Course flow diagram

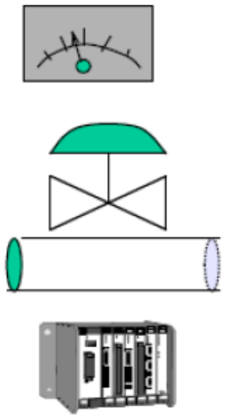
The main areas of automation

- Process automation
- Discreate automation
- Machine automation
- Building automation

Basics of automation technology

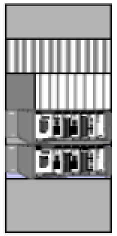
Hardware structure

- Meters, sensors
- Actuators, control devices
- Programmable logics (PLC)
- Process stations
- Supervisory control and data acquisition (SCADA)



Programming automation functions

- Logic programming
- Function block programming



Laboratory works and exercises

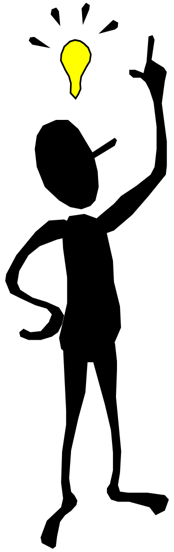
- Laboratory works
- returnable exercises

Basics of automation technology

Automation
Automation technology

Why automation?

- Management of industrial processes is easier
- Fast processes impossible to adjust by humans
- Processes that contain many control items can be managed only with automation
- Enables 24 h/day production
- Increases the quality of production and the uniformity of quality
- Automation system is TIRELESS



Source: Seppo Rantapuska

The main areas of automation

- Process automation
  - Continuous processes
  - Batch processes
  - Food industry, chemical industry, paper industry, power plants, etc.
- Discreate automation
  - Control different lines, conveyors, material flows
  - Handling, palleting, packing, automated warehouses
  - Robotos and manipulators
- Vehicles and machine automation
  - Machine tools, turning machine
  - Excavator, harvester
- Building automation
  - Heating, cooling, IV, lighting, monitoring and alarms, etc

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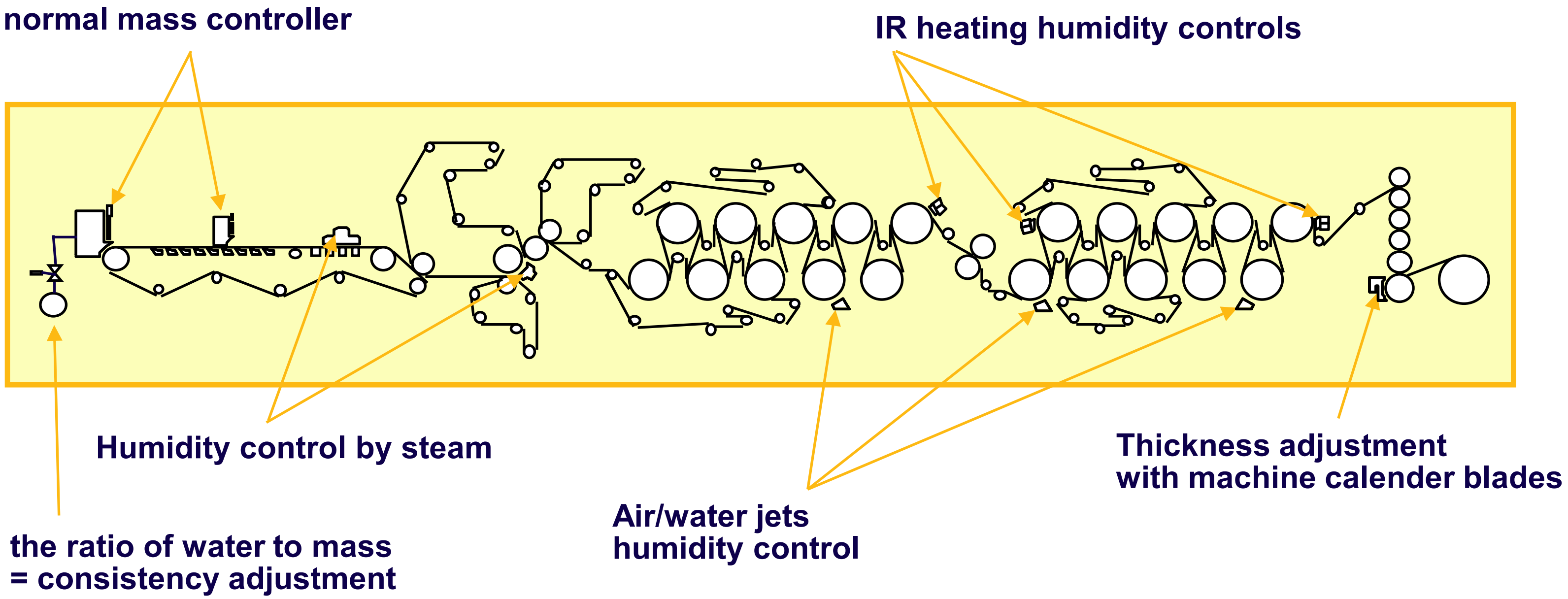
Process automation - continuous

- A continuous process, as the name implies, is continuous: raw materials are fed into it as a continuous supply, process steps are carried out, and at the end, a continuous stream of the output of that process or sub-process (material or, for example, energy in the case of power plants) is produced.



Valmet paper machine:
https://www.youtube.com/watch?v=sdWRovWKxU4
Stora Enso cardborad:
https://www.youtube.com/watch?v=41QVccYXoHE&t=408s

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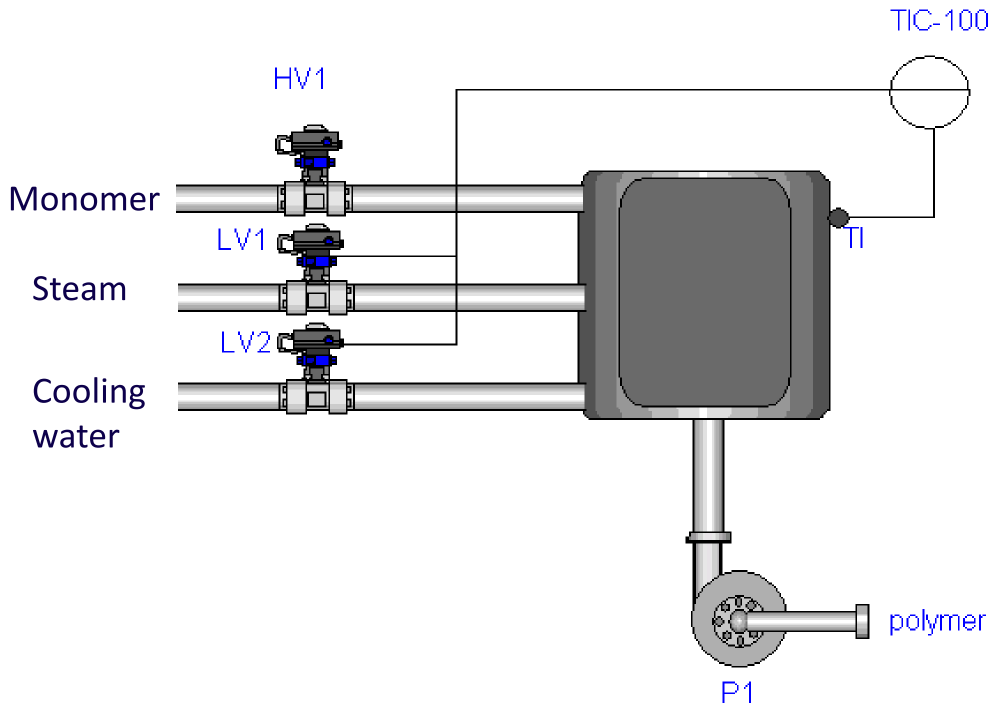
Process automation - batch

- In a batch process, the final product of a process or sub-process is manufactured one batch/portion at a time (not in a continuous stream).
- Stages of the batch process, for example:
  - Dispensing the raw material(s) into the reactor
  - Stir for a certain time at a certain temperature and pressure
  - Cool
  - Move the output of the reaction forward



- BatchChannel batch process animation:

https://www.youtube.com/watch?v=IlH-tSxlTRE

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Discrete automation

- In discrete automation (or production automation) solid piece-like products are moved. E.g.
    - In-plant logistics
    - Conveyors
    - Assembly
    - Robots
    - AGVs or mobile robots
    - Product packaging

ABB automated order picking system

https://new.abb.com/news/detail/90446/cstmr-meat-processing-company-uses-abb-automated-order-picking-system-to-speed-up-delivery-times-and-extend-product-shelf-life

JAMK O-ring demo:

https://www.youtube.com/watch?v=ucZBG2QLk98

Ramonedge machine tending :

https://www.youtube.com/watch?v=BaGp1tAETw0

Vähälä terminal, euro pallet handling:

https://www.youtube.com/watch?v=gPsDESSlOSs

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Machine automation

- For example, machining cells and machine tools:
- The machine tool contains a lot of precise motion control and tool changes
- In automatic machining cells, the robot serves the machine tool (machine tending)

Lethe + machine tool robotization

https://www.youtube.com/watch?v=q8WFAr84NzA

5-Axis Vertical Machining Center

https://www.youtube.com/watch?v=CqePrbeAQoM

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More about Jamk’s robotics

- Robotics by Jamk

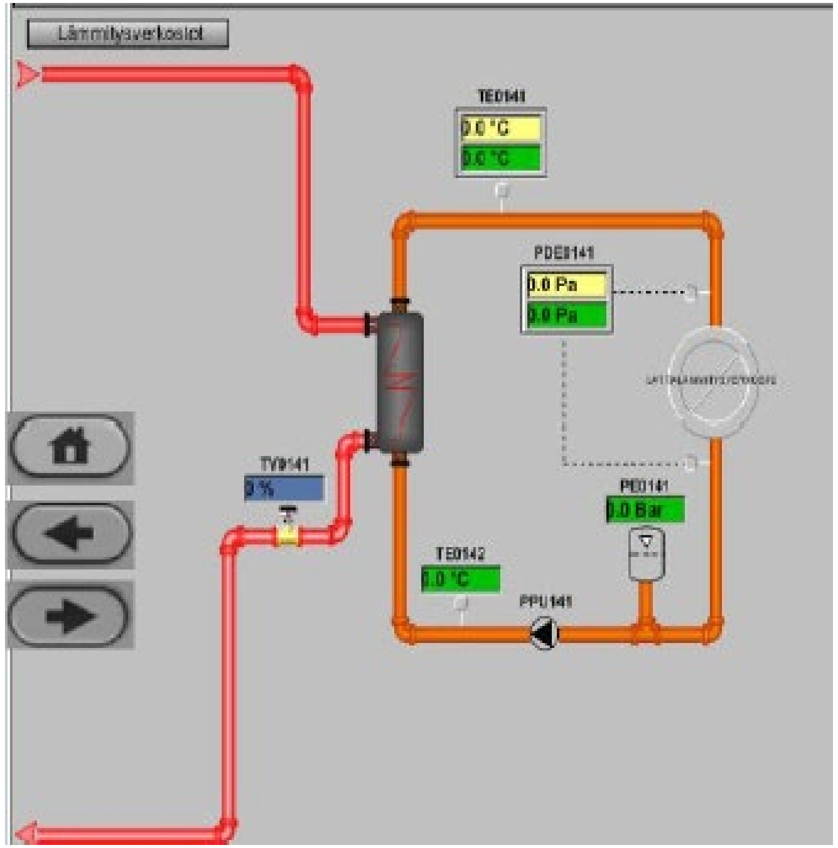
https://www.jamk.fi/fi/projekti/robotics-by-jamk

- Robotics by Jamk Youtube channel:

https://www.youtube.com/playlist?list=PLX7HAK4dyaDLK1A4REFXH3VbZ0ygYbVfh

Building automation

- Heating network control
- Ventilation control eg. Based on CO2 measurement
- Cooling
- Lighting control
- Measurements, monitoring, alarms



Work machine

- For example, forestry machines (harvesters)
- The driver controls the machine with controls connected to the automation system (e.g. joysticks, buttons/switches, touch screen) → The automation system implements accurate and fast "sub-level" controls and other functions of the machine based on the signals
  - Hydraulic frame steering, traction and hydraulic boom control
  - Automatic log cut-off to the correct dimensions
  - Length measurement sensor and frame thickness measurement at the harvester head
- Mining and rock solutions (Sandvik)

https://www.youtube.com/watch?v=difuW752jNU

https://www.youtube.com/watch?v=TdYpxF9-8Ec

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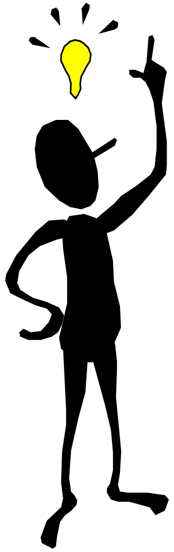
Automation
Automation technology

What is automation?

- automatic control and monitoring of industrial processes
- automation includes measurement, controls and adjustments
- Measuring devices, actuators, computers, SCADAs



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Automation system – Why?

- Previously controls, measurements and controls were separately
- In the automation system, all data is centralized under one system
- Helps process operators, faster and more accurate than humans
- Contains measurement and control data, performs automatic adjustments and on/off turns on actuators
- Devices can be from different suppliers
- Visually shows the user the status of the process through the user interface
- The user can change parameters and tuning values through the automation system centrally
- Give alerts and warnings
- Automatically switches the process to the correct mode in case of breakdowns (safety locks)
- Perform automatic operations of recipes and sequences , eg. variety change, change of recipe
- provides information about the process to other systems (e.g. production control systems)
- provide information on the condition of the equipment, the need for maintenance and the quality of production
- Provides history data of the process (shift reports, daily reports, weekly reports)

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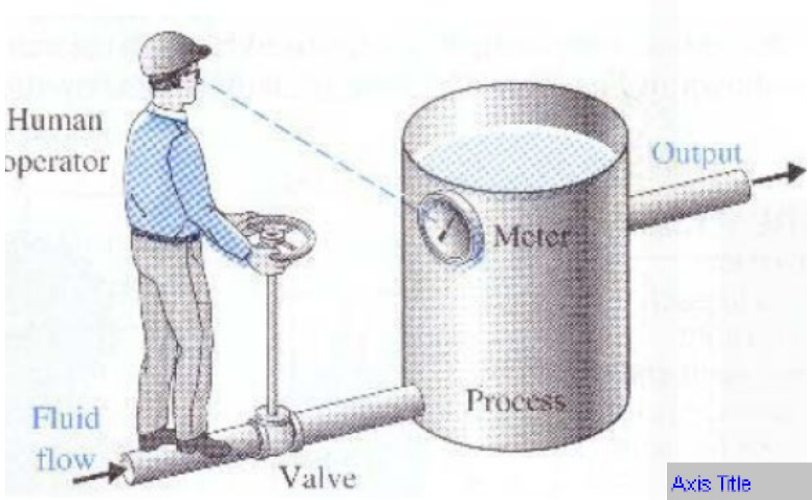
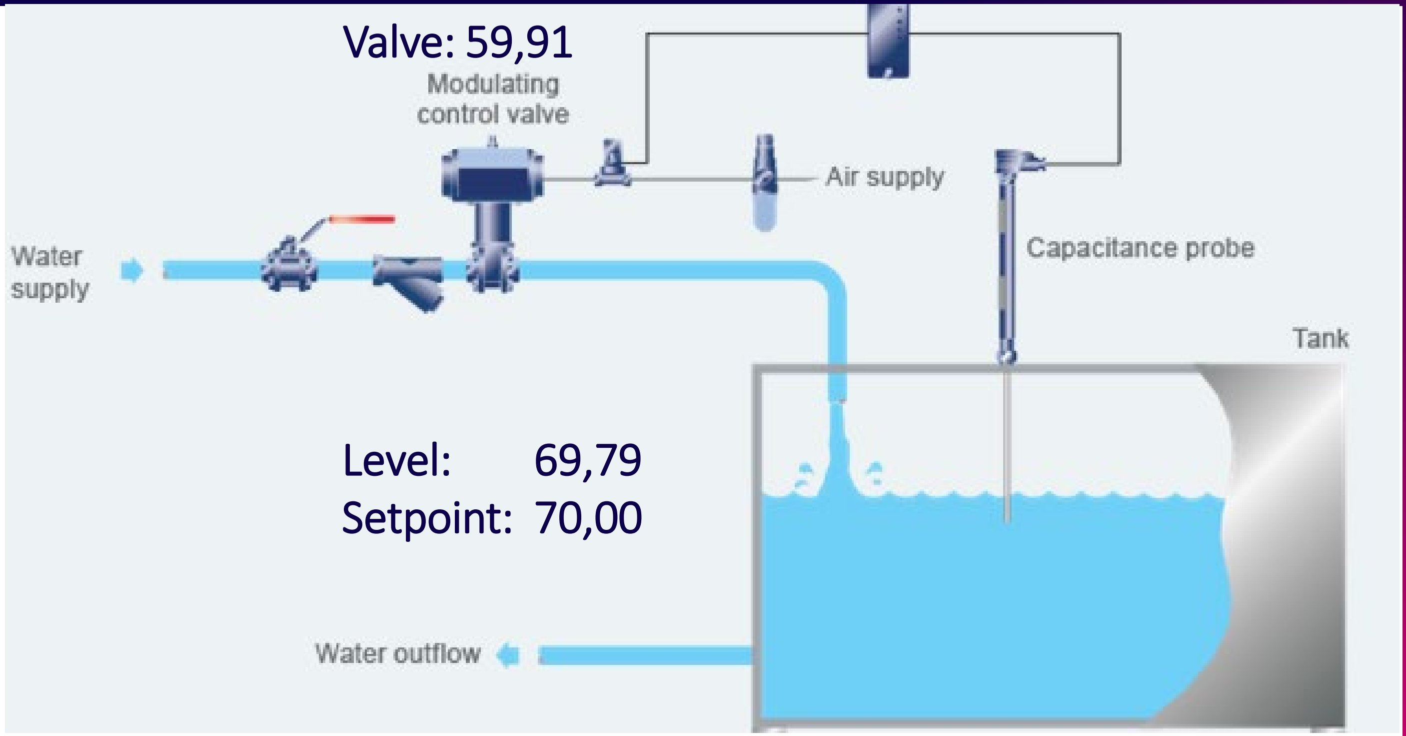
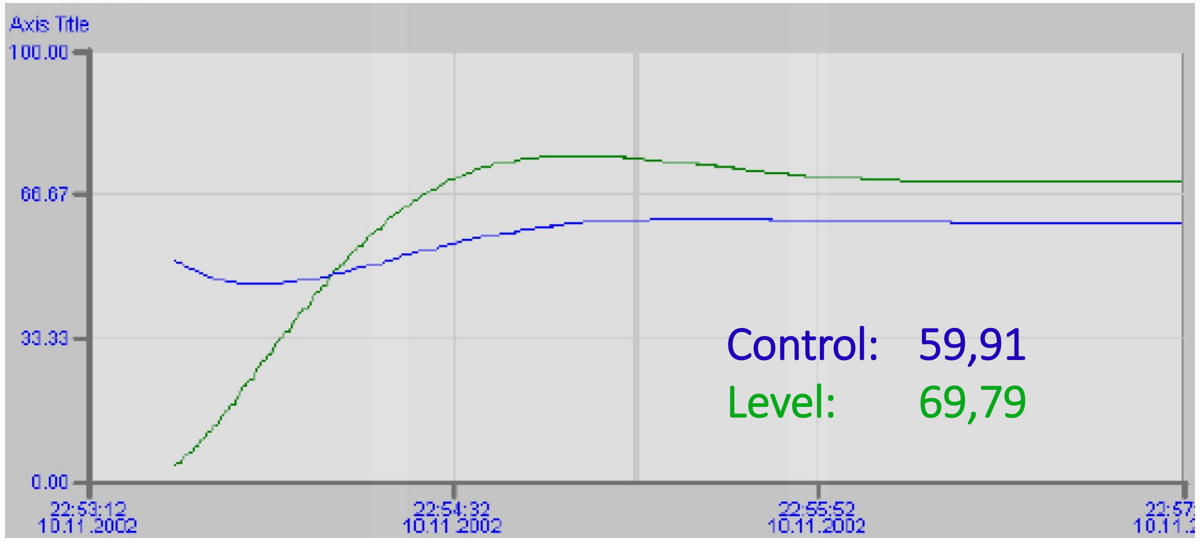
Automatic control:

- Process industry: level control of the tank







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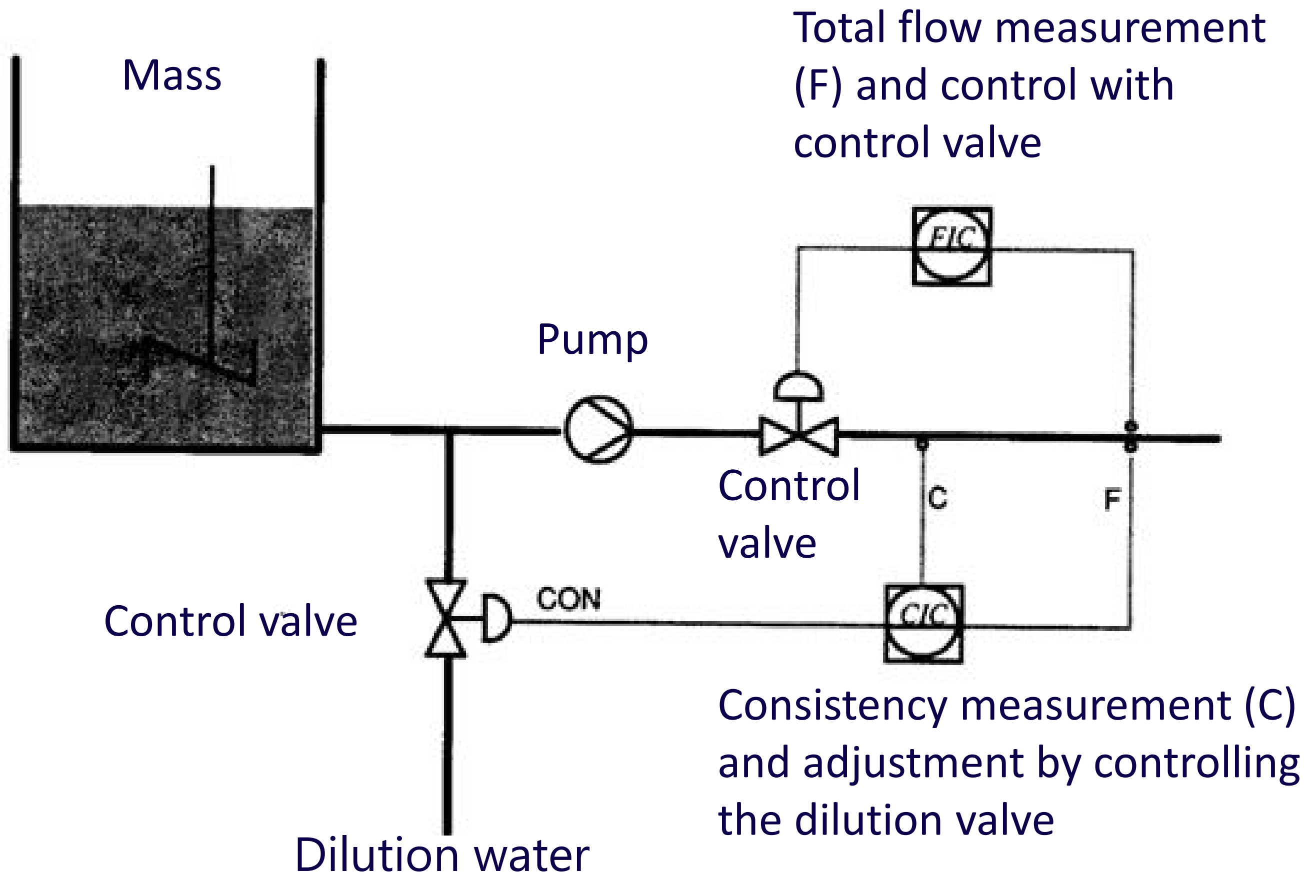
Example diagram (continuous adjustment: consistency adjustment)



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